Nernst-Planck equation

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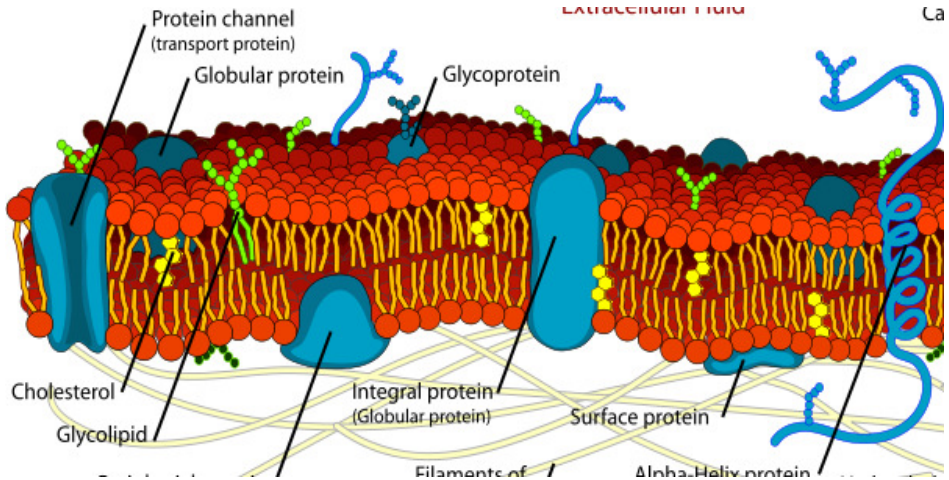
Biomatematica 2

October 19, 2020, Milan, Italy

The Cellular Membrane



- The cellular membrane is a lipid bilayer in which are immersed proteins
- Proteins behave as channels, allowing the flow of ions through the membrane
- Ion concentrations gradients are maintained between the two sides of the membrane
- The concentrations gradients produce a difference of potential through the membrane, the transmembrane potential
- The energy spent to maintain the concentrations gradients is provided by ATP (Adenosintriphosphat)

The Sodium-Potassium pump

The **NaCl** e **KCl** salts dissolved in the water : **Na^+**, **K^+**, **Cl^-**

Pump
out 3 Na^+
inside 2 K^+

Energy Store
biochemical
ionic batteries

biochemical
process
Ionic Pump Na^+-K^+

$[Na^+]_i < [Na^+]_e$ at the equilibrium the Na^+ current is directed from the intracellular space $\rightarrow$ toward the extracellular space pot. $v_{Na} = +50mV$

$[K^+]_i > [K^+]_e$ the K^+ current is directed from extracellular space $\rightarrow$ intracellular space pot. $v_K = -77mV$

The Sodium-Potassium pump II

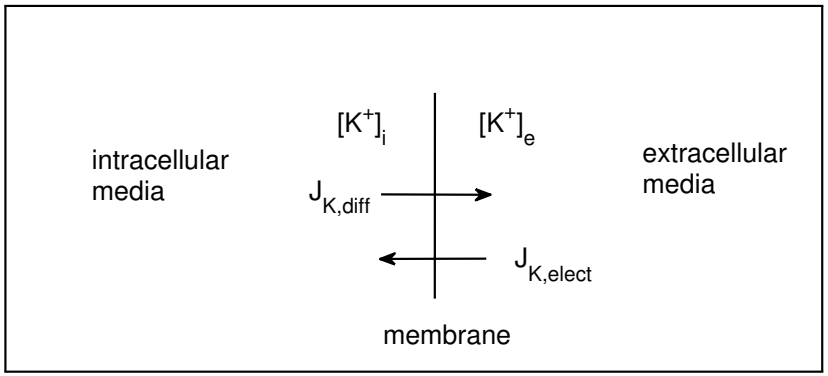
At the equilibrium this biochemical pump maintains a potential jump between the intra and extracellular space of $v_{rest} = -65mV$

Na/K-ATP-ase sodium-potassium pump: ions transport against gradients . Active transport i.e. with energy consuming transport with conversion of ATP $\rightarrow$ ADP

$$\boxed{\mathrm{ATP} + 3\,[\mathrm{Na}^+]_i + 2\,[\mathrm{K}^+]_e \rightarrow \mathrm{ADP} + \mathrm{P}_i + 3\,[\mathrm{Na}^+]_e + 2\,[\mathrm{K}^+]_i}$$

One-ion flux

Suppose that the membrane is permeable only to the potassium (K^+) ion with valence $+1$.



Ionic Membrane Current

$J_{K,diff}$:= diffusion K^+ flux

$J_{K,elect}$:= electric field K^+ flux

J_K := total K^+ flux

$$\boxed{J_K = J_{K,diff} + J_{K,elect} \ (mol/cm^2 s)}$$

The diffusion flux $J_{K,diff}$ and the electric flux $J_{K,elect}$

$$\boxed{J_{K,diff} = -D_K \nabla c_K}$$

Fick's law: diffusion motion

D_K := diffusion coefficient (cm^2/s)
c_K := K^+ concentration (mol/cm^3)

$$\boxed{J_{K,elect} = -\frac{z_K}{|z_K|}\mu_K c_K \nabla u}$$

Planck's equation:
electronic motion of ions

u := electric potential (mV)
μ_K := mobility of the ion (cm^2/mVs)
z_K := valence of the ion $\quad \frac{z_K}{|z_K|} = \begin{cases} +1 & \text{positive ions} \\ -1 & \text{negative ions} \end{cases}$

[concentration]=0 $\rightarrow \quad J_{k,diff} = 0$
[potential]=0 $\rightarrow \quad J_{K,elect} = 0$

The Nernst-Planck equation

The relationship between the diffusion coefficient D_K and the ion mobility μ_K is given by

$$\boxed{D_K = \frac{\mu_K RT}{|z_K| F}} \qquad \textit{Einstein} \ (1905)$$

$R :=$ gas constant
$T :=$ absolute temperature
$F :=$ Faraday constant

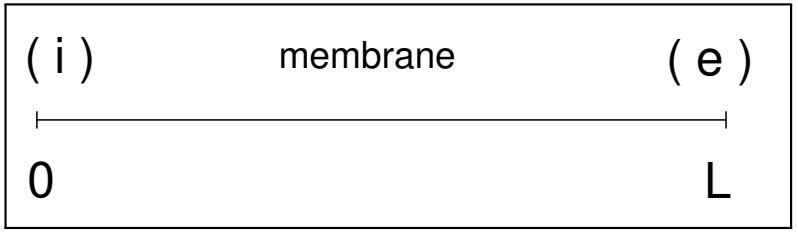
Hence it follows that the total ionic flux J_K is

$$J_K = J_{K,diff} + J_{K,elect} = -D_K \nabla c_K - \frac{z_K}{|z_K|} \mu_K c_K \nabla u$$

$$\boxed{J_K = -D_K \left(\nabla c_K + \frac{z_K F}{RT} c_K \nabla u \right)} \qquad \textit{Nernst-Planck equation}$$

The Nernst potential

At thermal equilibrium: every local process and its reverse proceed at the same rate. $J_K = 0 \quad \Rightarrow \quad \boxed{\nabla c_K = -\frac{z_K F}{RT} c_K \nabla u}$



$$\frac{1}{c_K}\frac{dc_K}{dx} = -\frac{z_K F}{RT}\frac{du}{dx}$$

$$\int_0^L \frac{1}{c_K}\frac{dc_K}{dx}\,dx = -\frac{z_K F}{RT}\int_0^L \frac{du}{dx}\,dx$$

$$\Downarrow$$

$$\log\left(\frac{c_K(L)}{c_K(0)}\right) = -\frac{z_K F}{RT}\left(u(L) - u(0)\right)$$

$$\Downarrow$$

$$\boxed{v_K := u_i - u_e = -\frac{RT}{z_K F}\log\left(\frac{c_K^i}{c_K^e}\right)}$$

Nernst potential

Table of Nernst potentials

	extracellular concentration (mM)	intracellular concentration (mM)	Nernst potential (mV)
Na^+	145	15	60
Cl^-	100	5	-80
K^+	4.5	160	-95
Ca^{2+}	1.8	1e-4	130
H^+	1e-4	2e-4	-18

Table: Intra- and extracellular concentration and Nernst potential values for ventricular myocytes.

The Goldman-Hodgkin-Katz (GHK) current-voltage relation I

Suppose the electric field constant through the membrane, i.e.

$$\boxed{\frac{du}{dx} = -\frac{V}{L}} \qquad \text{where } V = u(0) - u(L) = u_i - u_e.$$

Moreover at the steady state the flux J_K must be constant.
In this case the Nernst-Planck equation is an ODE for c_K,

$$\boxed{\begin{gathered} \frac{dc_K}{dx} - \frac{z_K F V}{RTL} c_K + \frac{J_K}{D_K} = 0 \\ c_K(0) = c_K^i \quad c_K(L) = c_K^e \end{gathered}}$$

whose solution is

$$\boxed{c_K(x) = -\frac{J_K RTL}{D_K z_K VF}\left(1 - \exp\left(\frac{z_K VF}{RTL}x\right)\right) + c_K^i \exp\left(\frac{z_K VF}{RTL}x\right).}$$

The Goldman-Hodgkin-Katz (GHK) current-voltage relation II

To satisfy the BC $c_K(L) = c_K^e$, it must be that

$$\boxed{J_K = \frac{D_K z_K F V}{LRT} \frac{c_K^i - c_K^e \exp\left(\frac{-z_K V F}{RT}\right)}{1 - \exp\left(\frac{-z_K V F}{RT}\right)}.}$$

Defining the electrical current density $I_K := z_K F J_K$, we have

$$\boxed{I_K = P_K \frac{z_K^2 F^2}{RT} V \frac{c_K^i - c_K^e \exp\left(\frac{-z_K V F}{RT}\right)}{1 - \exp\left(\frac{-z_K V F}{RT}\right)}} \qquad \textit{GHK equation}$$

where $P_K = \frac{D_K}{L}$ is the permeability of the membrane to the ion K^+.

We remark that: $I_K = 0$ **if** $V = v_K$ the equilibrium (reversal) Nernst potential.